

الجمهورية العربية السورية
وزارة التربية

English for Starters

Vocational Stream

Industrial Grade 12

Computer and Electronic Technology

Activity Book



٢٠١٨ - ٢٠١٩ م
المؤسسة العامة للطباعة

Contents

Unit	Title	Page
1	Optical Storage	3
2	Digital Cameras	6
3	Power Supply	9
Test 1		12
4	Sensors	14
5	Maintenance of Home Appliances	17
6	Installation of Lifts and Escalators	20
Test 2		23
7	Inside the System	25
8	Robots	28
9	X-rays	31
Test 3		34
10	Automatic Control	36
11	Electric Power Station	39
12	Latest Discoveries	42
Test 4		45
Glossary		47

1 Optical Storage

Exercise 1 Vocabulary

Fill in the blanks with the appropriate word from the box.

player optical storage laser beam
CD information digital bumps

- a** The best method to store our data and keep it safe is using an device.
b I have recorded my favourite music tracks on a
c Our company's data is very valuable. Why don't we store it on DVDs?
d If you look at the surface of a CD, you will find a spiral track that is formed by tiny
e A reads the data that has been recorded on the CD.
f I need to know some in order to know which optical storage device I have to use.
g I got a new CD for my birthday. It was just what I wanted!



Exercise 2 Matching

Match the words in column A to their functions in column B.

Column A	Column B
1 circuit board	a reads the data
2 rail	b provides power
3 spindle	c contains electrical connections
4 laser lens	d helps the motor to move
5 motor	e holds the CD

Exercise 3 Grammar

Use the words in brackets to rewrite the following structures. You may change the tense used or the order of the phrases.

- 1 Have you forgotten to call the technician? (... yesterday?)
- 2 You don't have to speak to the manager because she has already phoned him.
(She has ... so ...)
- 3 Is it natural for a CD to take plenty of time to be recorded? (Does ... ?)
- 4 He is using modern equipment to make thousands of DVDs in a very short period of time, isn't he? (Is he ... ?)
- 5 The factory finished manufacturing the CDs before the customer cancelled his order.
(When the customer ...)
- 6 The package arrived when the employees were having lunch. (The employees were ...)

Exercise 4 Reading

Read the following text about some types of DVD and then answer the questions that follow.

There are many types of Digital Versatile Discs, or DVDs. A DVD-R can only be used to record data one time and cannot be re-written. A DVD-RW, however, can be re-written many times, which means that it can be used for recording more than once. Both the DVD-R and the DVD-RW have a storage capacity of 4.7Gb. Another type of DVD is the DVD-R Dual Layer (DL) disc, which is the most impressive disc as it has 8.5 Gb of storage. This means that it can store up to four hours of video or 120 hours of MP3 audio. 'Dual Layer' means that data can be recorded on one side of the disc while the other side is not affected. However, the disadvantage of the DVD-R DL disc is that it is more expensive than the standard DVD-R and DVD-RW.

Questions

- 1 How much storage capacity do the DVD-R and DVD-RW have?
- 2 What is the benefit of the DVD-RW?
- 3 What is the disadvantage of the DVD-R DL?
- 4 Find words from the text that have the same meaning as:
 - a to write over existing data
 - b double
 - c touched
 - d costing more money

Exercise 5 Speaking

Work with a partner and hold a discussion about the importance of optical storage in our daily lives. Use the phrases below while discussing and then share your answers with the class.

- There are so many methods of storing ...
- We use DVDs to ...
- Lasers can be found in ...
- (Type of storage) can be used to ...
- For example, optical storage helps ...

Exercise 6 Writing

Write a set of instructions in the form of a short paragraph to explain how to record data on a CD. You can use the words below.

burn / insert / data / software / select / drag /
wait / buy / open / audio / choose

.....

.....

.....

.....

.....

.....

.....



2 Digital Cameras

Exercise 1 Vocabulary

Put the digital camera key words in the box in the right place.

bits and bytes pixels semiconductor lens electronically

The quality of the photographs in a digital camera depends upon the camera resolution, which is determined by **(1)**

The built-in computer in a digital camera allows images to be recorded **(2)**

The **(3)** is a piece of glass which is an integral part of any camera and determines the quality of the photographs.

The language that the computer recognises is represented in **(4)**

A digital image is created when light is focused onto a **(5)**

Exercise 2 Vocabulary

Choose the correct options for each of the sentences below.

1 A is a device that measures the speed of moving objects, such as vehicles. This device is used to make sure drivers do not exceed the specified

a speed limit **b** speed gun

2 A specifies the distance of a car from the speed gun. It is projected from a small part of the speed gun called the

a laser transmitter **b** laser beam

3 The allows the police to check the speed of a car and decide whether the speed limit is broken, whereas a allows the driver to know and control the speed of their car.

a telescopic sight **b** speedometer

Exercise 3 Grammar

A Choose the correct words to complete these sentences.

- I the photo when you're all ready.
a will take b take c am taking
- you connect your camera to your laptop, the photos will start to download.
a As soon as b Before c Until
- When you fully charge your camera, it for longer.
a will work b worked c was going to work
- You to read the instruction manual you first operate your camera.
a will have, before b have, after c will have, until
- If you a new digital camera, it will help you keep a lot of memories.
a will use b used c use

B Write six sentences about digital cameras using future time clauses.

Exercise 4 Reading

Read the text and answer the following questions.

Digital Camera Cleaning and Maintenance



Lenses

The glass is a very sensitive part of the camera. This is why it is expensive to repair or replace. Not any tissue or alcohol can be used to wipe the lenses. There are some fine tissues and alcohol solutions especially for cleaning cameras. But before that, begin by using a blower brush to remove any dust from the front and the back elements.

Camera Sensor

Turn on the camera and set it to manual with a thirty-second exposure to make sure the shutter doesn't close while you're cleaning the sensor. Once the shutter is open, hold the camera up so that the sensor is facing down, and use the blower to blow any dust away from the sensor. Do not use the brush; blowing the dust away is enough to clean the sensor.

Outer Casing

The external parts of the camera have to be clean in order to prevent the dirt on the outside to make its way inside. That's why, it is better to do the outer parts before the inner parts. Use the blower brush first to clean the outside and then a quick rub with a lens cloth or dust cloth.

Questions

- Find two synonyms and two antonyms in the text.
- Why do you think the blower should be used without the brush while cleaning the sensor?
- How should the outer casing be cleaned? Why?

Exercise 5 Speaking

The following short texts describe some features of the digital camera. Read them and match them to the appropriate titles in the box, then search the Internet for more features to present to your teacher and classmates.



Image stabilisation Face detection Focus
Shooting modes Exposure modes Zoom lenses

- 1 It manages the shutter speed according to the available light.
- 2 This type of lens, which is actually made up of several different lenses or lens elements, allows you to vary the focal length. This provides you with flexibility in framing shots and closes the distance between you and your subject. Optical zooms are much better than digital zooms, which merely magnify the centre of the frame without actually increasing picture detail.
- 3 More and more cameras, including many with powerful lenses, now come with it, a device that compensates for handheld camera shake.
- 4 This digital camera feature attempts to find a face in the image to set focus, exposure, and colour balance so that faces appear in focus and well exposed.
- 5 Digital cameras automatically adjust the focus of the lens with autofocus features.
- 6 Most cameras have three options for shooting still images: single image, burst mode, and self-timer. The burst mode allows you to take several shots quickly. In some cameras this could go up to several shots per second. The self-timer mode provides a delay between the moment the shutter button is pressed and the photo is captured.

Exercise 6 Writing

Use the following phrases to write a paragraph on how to buy a digital camera.

digital photography / higher-resolution images / sophisticated but easy-to-use controls / higher optical zoom levels / shooting video / easier than ever / it isn't all about megapixels / size, weight, and design / storage / memory card / wireless

.....

.....

.....

.....

.....

.....

.....

3 Power Supply

Exercise 1 Taxonomic skills

Rearrange the following steps in the correct order.

- The regulation stage is when a regulator provides constant and regular output of power.
- The smoothing stage is when a capacitor regulates the amount of electricity to the required level.
- The rectification stage is when a rectifier converts AC to DC.
- There are three main stages for converting AC to DC power.
- The diodes in the rectifier allow electricity to pass one way and not the other.

Exercise 2 Grammar

Complete these sentences by filling in the blanks with the correct relative pronoun.

- 1 A power supply, is installed inside a computer, is essential to the functioning of a computer's internal hardware.
- 2 At the back of a power supply, you can find a fan opening, air is blown out of the back of the computer case.
- 3 An electronic technician, helps create, install and fix electronic equipment like computers, is the person needed for the job.
- 4 This electronic technology student, portfolio has impressed our company, will be doing an internship starting next week.



Exercise 3 Reading

Building an efficient power supply is necessary in order to prevent energy loss. This is important for the environment and also for the user. High efficiency in power supplies is especially important for portable devices to make sure the battery life lasts. Most power supplies these days work at 90% efficiency levels. One way to ensure efficiency is to make the power supply unit as small as possible. A simple design is also desirable to keep the costs of manufacturing low. In addition, the power supply needs to be well-designed so that it prevents overheating and energy loss. With a more effective power supply, less heat is produced, so less heat needs to be removed. Therefore, the unit can be smaller as it will not require a cooling fan or heat sink. Designs which allow air to flow around the parts that will get hot are good. Power supply designers are constantly working on producing a product that converts AC to DC power efficiently, has an accurate output voltage and reduces heat. These will save you money on energy, but the product will be more expensive to buy.

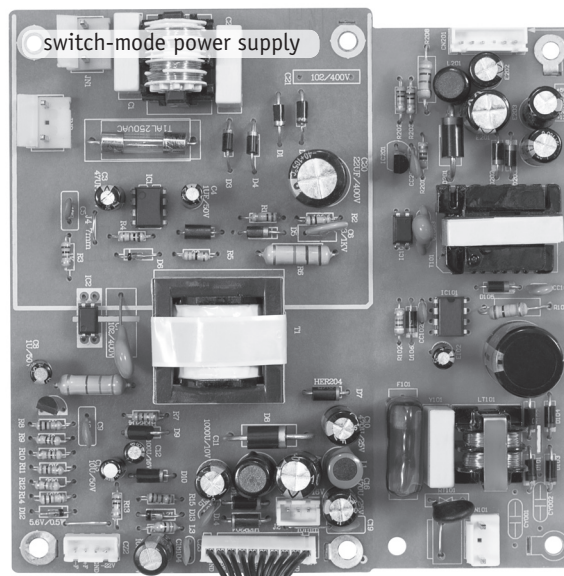
Switch-mode power supplies are very effective as they are small in size and weigh less than other power supplies; they do not use a transformer to convert AC to DC power. They also prevent energy loss as they do not lose power.

There are also new types of rectifiers that make power supplies more efficient. These are called synchronous rectifiers. The quality of the semiconductors is also important. New semiconductors are being produced to help improve efficiency.

By combining the best design technologies with the latest parts, it is possible to make efficient power supplies. Future designs will look at both which the materials to use and how to put them together in order to increase the efficiency of power supplies.

Questions

- 1 Give the text a suitable title.
- 2 Are the following statements true or false?
 - a Building an efficient power supply is important for the environment.
 - b The smaller the power supply gets, the more efficient it becomes.
 - c Switch-mode power supplies waste a lot of energy.
 - d Synchronous rectifiers are a traditional design of power supplies.
- 3 List the factors from the text that can make a difference to the efficiency of a power supply.
- 4 Why is the design of the power supply important?



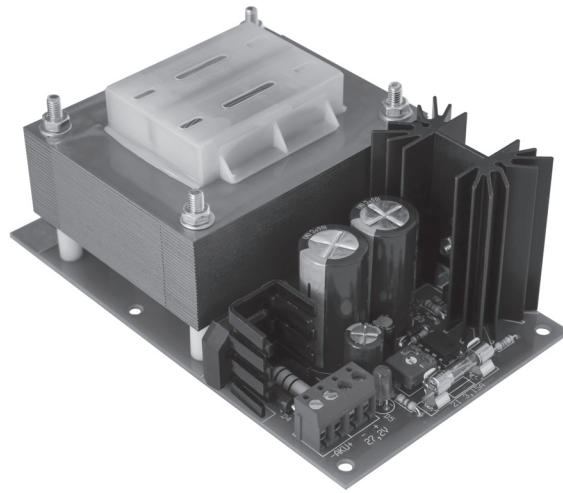
Exercise 4 Speaking

Work in groups. Do some research about the different power supplies being used today. Choose one type and make an oral presentation about it.

Exercise 5 Writing

Write a paragraph about building an AC power supply using the steps below.

The major steps in constructing an AC power supply are: planning layout -----> installing transformer -----> soldering power supply -----> preparing diode bridge -----> soldering wires -----> attaching capacitor -----> connecting LED light -----> connecting output cable



Test 1

Part One: Reading comprehension (35 marks)

I Read the text and then answer the following questions.

Features of Power Supplies

A power supply **requires** certain operational features in order to protect it. These include: overcurrent, overtemperature, overvoltage, soft start, power factor correction and electromagnetic compatibility.

- It is important that the output current of a power supply does not **exceed** the maximum allowed current. For this reason, power supplies require built-in overcurrent protection to prevent damage to the power supply and its cables.
- Excessive temperature can also damage the power supply. Therefore, most units have a temperature sensor that **disables** the supply if the temperature goes above a specific value.
- Overvoltage protection is also **critical** when considering power supply protection. Most power supplies will switch off automatically if the output voltage is excessive.
- When the power is initially switched on, a rush of current can occur which may affect the output voltage of the supply. To avoid this, a soft-start circuit can be installed which limits the first rush of current and then regulates the current to normal output.
- Power factor correction circuits ensure that AC-DC power supplies are efficient by modifying the voltage and current on the power line.
- Finally, electromagnetic compatibility is an important feature as it minimises electromagnetic interference.

Questions

- 1 What are the features that can be used to protect power supplies?
- 2 Why is it important for power supplies to have the features mentioned in the text?
- 3 How does a temperature sensor protect the power supply?
- 4 Explain the words in bold.

Part Two: Grammar and structure (60 marks)

II Fill the text with words from the list.

coils magnetic field power supply
input voltage output voltage
circuits electromagnetic induction
electrical energy

A transformer is an important feature of an AC **(1)** The function of the transformer is to step down the **(2)** to a lower **(3)** A transformer transfers **(4)** between **(5)** Energy is transferred by **(6)** A transformer has a primary coil which is connected to an AC supply. Current passes through the coil which causes a change in the **(7)** This in turn changes the voltage in a secondary coil. It is important to note that there is no electrical connection between the primary and the secondary **(8)** The latter induces alternating current in the circuit it is attached to.

4 Sensors

Exercise I Vocabulary

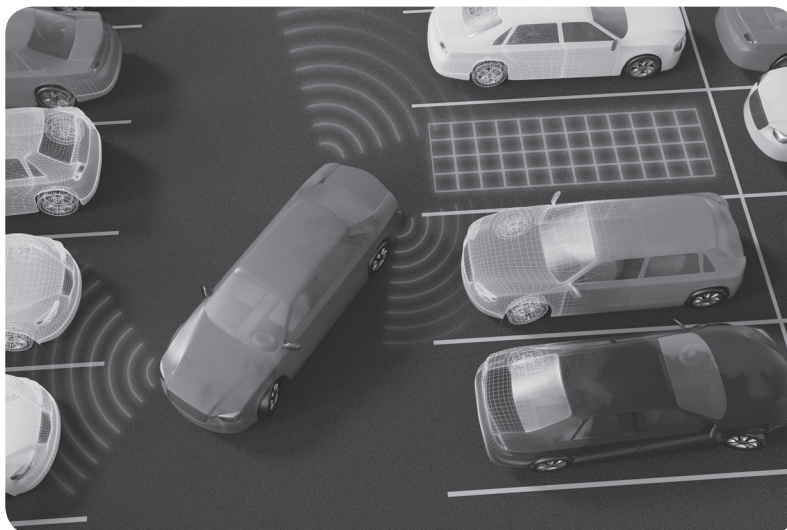
A Complete these sentences by filling in the blanks with the correct words from the box.

bumper installed sensor parking

- 1 A car can help a driver to park safely without having to see.
- 2 a vehicle can be made much easier now that we have technology that helps us.
- 3 I had my sensor last week by a professional mechanic and technician.
- 4 Most car sensors are placed on the car's to ensure it can detect areas not visible to the driver.

B Use each of the following words and phrases in a meaningful sentence.

- 1 control box
- 2 parking
- 3 instruction manual
- 4 boot



Exercise 2 Dialogue

Work in pairs. Refer to the dialogue on page 20 of your Students' Book. Student A plays the role of a mechanic. Student B is a customer who wants to have parking sensors installed on his car.

A

B

A

B

A

B

Exercise 3 Grammar

A Rewrite the following sentences in the passive voice. Use an appropriate auxiliary or a modal verb to make meaningful sentences.

1 Two or four sensors / install / on / car's bumper.

.....

2 Yesterday, / my car sensors / repair / due to the road accident.

.....

3 This weekend, / my car / check / a mechanic.

.....

4 Sensors / install / according to / instruction manual.

.....

5 In the future, / much more advanced parking sensors / manufacture.

.....

6 Sensors / test / before driving / to make sure / they / install / correctly.

.....

B Put these sentences in the passive form.

1 You should test the connections and make sure the control box is working properly.

.....

2 Before you install the sensors, you have to choose a location for their placement.

.....

3 You must disconnect the car battery to prevent yourself from getting injured when you install the sensors.

.....

4 You can install a warning siren in the trunk to warn you if you're reversing too close to a car.

.....

Exercise 4 Reading

Read the passage below about temperature sensors and answer the questions that follow.

How Temperature Sensors Work

There are two types of temperature sensors. A contact temperature sensor is able to measure the temperature of something that it touches. An example of this would be the digital thermometer you use to see if you have fever when you are unwell. In electrical currents, thermometers measure changes in resistance. This measurement is converted from electrical resistance to temperature through a microchip. A non-contact temperature sensor, however, does not need to touch the object to measure its temperature. This means it can measure the temperature of something that is not close by. It works by measuring the thermal radiation that comes from an object. It is useful for measuring the temperature of moving objects, for large areas and for objects that are far away and difficult to reach, such as outer space.



Questions

- 1 What are the two types of temperature sensors called?
- 2 What does the contact thermometer measure?
- 3 How are non-contact thermometers useful?
- 4 Find words from the text that have the same meaning as these words / phrases:
 - a raised temperature in the body
 - b alterations
 - c relating to heat
 - d size, length or amount of something

Exercise 5 Speaking

Carry out a conversation with a partner about the usefulness of sensors in our daily lives. Refer back to the exercises in this unit for ideas.

Exercise 6 Writing

Write a short paragraph describing what kind of appliances in your home use sensors. Give your paragraph a suitable title.

5 Maintenance of Home Appliances

Exercise 1 Vocabulary

Choose three common faults (1-10) for each device (a-j). Some devices have more than three answers.

Device	Fault	Choose from this list:
a computer		1 The paper doesn't come out.
b television		2 The cover doesn't open.
c radio		3 The sound goes quiet.
d car		4 The fan doesn't turn.
e motorbike		5 The engine doesn't start.
f video recorder		6 The screen goes black.
g fax		7 The tape doesn't turn.
h mobile phone		8 The tyre has a puncture.
i camera		9 The antenna is twisted.
j air conditioner		10 The power doesn't come on.

Exercise 2 Dialogues

Write questions and answers based on the chart above.

Examples:

A: What's the problem?

B: My computer is broken.

A: What's wrong with it?

B: The power doesn't start.



Exercise 3 Grammar

Rewrite the following questions in the reported speech form.

- 1 Khaled: "How can I learn about maintaining home appliances?"
- 2 Ruba: "How do I keep my appliances safe during thunderstorms?"
- 3 The manager: "Who can take care of my broken TV?"
- 4 Amer: "Can you finish restoring my laptop by tomorrow?"
- 5 Hani: "Is it possible to identify the reason why the radio keeps breaking down?"

Exercise 4 Reading

Read the following, then answer the questions.

Mobile phone troubleshooting guide		
Problem	Probable cause	Possible solution
When the phone is switched on, there is no power.	The battery may not have been recharged, and it could have gone flat.	Recharge it immediately.
The power is on, but the other person's voice can't be heard.	The phone may not be able to transmit or receive a signal when obstructed by physical barriers (such as concrete walls).	Go outside or to a room where the signal can be received.

Fax to: Telco Phone Co.
From: Mr Ali Gaafar

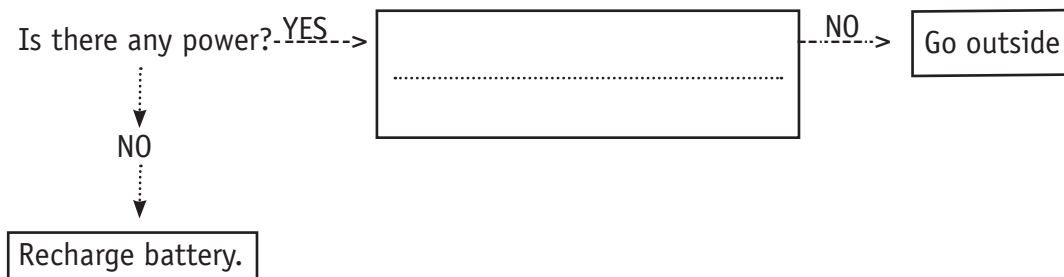
Dear Sir

I wish to complain about the mobile phone that **I** bought from **you** last week.

After I switch on the phone, I key in the number. The phone rings, but when the other person replies, I cannot hear him clearly.

Questions

- 1 What do the words in bold refer to?
 - 2 What barrier can stop a mobile phone from sending a signal?
 - 3 Which structure **A**, **B** or **C** should go into the flow chart box below?
- A** Can the other speaker be heard? **B** There is some power. **C** Has the battery been recharged?



- 4 Where do you think Ali Gaafar must have telephoned his friend from? Give a reason for your answer.

Exercise 5 Situations

What would you say in the following situations?

- a** You work in an electrical repair shop. You want to tell a customer that you have not been able to repair his VCR yet.
- b** A customer writes to your repair shop to complain. He says that his radio broke down after you repaired it. Write a reply in which you tell him that your shop will repair his radio free of charge.
- c** A customer enters the shop and gives his radio to you. You take it, but then you accidentally drop it on the counter.

Exercise 6 Writing

Write a short paragraph of five sentences to describe the computer mouse in the picture below.



.....

.....

.....

.....

.....

.....

6 Installation of Lifts and Escalators

Exercise 1 Vocabulary

Choose the answer that best completes the sentence.

- 1 The of the lift can't carry more than four people.
a cabin b clutch c cable d cord
- 2 lifts involve the use of a jack and a certain type of fluid.
a Traction b Pump c Lift d Hydraulic
- 3 The cabin of a lift is held up with a motorised
a clutch b cable c wire d connection
- 4 The should be the same shape as the lift cabin so that it can pass up and down through it.
a shoot b shuttle c shaft d shade
- 5 Press the number of the floor you want on the and the elevator will take you to that floor.
a control panel b board c button control d control

Exercise 2 Grammar

A Rewrite the sentences using the phrases in brackets to give the same meaning.

- 1 "I think that this escalator is not well installed." (He does not think ...)
- 2 "The public might not be aware that the lifts are still out of service." (I don't believe ...)
- 3 "Installing escalators will not reduce the number of customers who will use the stairs." (I don't expect that ...)
- 4 "Even though we have the best experts on installing lifts, we will still face maintenance problems and complaints from customers." (I believe that even though ...)



B Identify the functions in the following reported sentences.

e.g. Hani suggested going for a walk at the weekend.*suggesting*.....

- 1 Tina was not advised to take the lift when the fire alarm went off.
- 2 Dalia didn't warn Tarek about the consequences of overloading the lift.
- 3 Ramzi's parents never encouraged him to become a mechanical engineer.
- 4 The teacher decided to stop reminding Lama of her low grades.
- 5 Lama requested help from her teacher.

Exercise 3 Reading

Read the text and answer the questions.

Control Panels

When installing a lift, it is important to give special attention to the control panel. This panel is set with buttons that indicate the floors that the cabin can stop at. With the simple press of a button, the passenger can control what floor he / she wants the lift to go to. For a user-friendly control panel, the installer must consider a few basic points. This panel should provide the available choices the passenger can make. Then it should be able to allow the individual to make his choice. And lastly, the control panel should be able to confirm the choice made with a certain type of feedback. Taking these points into consideration will help in installing a user-friendly control panel for a lift.

Questions:

- 1 What is the main idea of the text?
- 2 Who do you think this text is written for?
- 3 What are some points to consider when installing a control panel?
- 4 According to the text, what is a control panel?



Exercise 4 Speaking

Discuss these questions.

- 1 Would you ever consider installing a lift? Why or why not?
- 2 What type of educational background would a lift installer need? Explain.
- 3 Which type of lift do you prefer: traction or hydraulic? Why?

Exercise 5 Writing

Choose one of the following types of lifts and write a paragraph in which you state its advantages and disadvantages. You can refer to the lesson in the Students' Book for ideas.

- Hydraulic lifts
- Traction lifts



Test 2

Part One: Reading comprehension (35 marks)

I Read the text then answer the following questions.

People who are responsible for the installation and maintenance of car sensor systems are called electrical automotive engineers. One of **their** tasks includes ensuring the operational controls such as checking that sensors are working correctly. Automotive electronics is an important aspect of automotive engineering. Without electronic controls, cars would not be able to meet the standards of modern safety requirements. Sensors play an important part in ensuring that cars operate safely and efficiently.

Nowadays with the invention of computerised vehicles, cars are being manufactured with a variety of different sensors. For example, there is a sensor that measures the pressure at which the oil pump is pumping engine oil. This sensor protects the engine from damage that would occur if the oil pressure is too low. The oxygen sensor controls ensure the engine runs efficiently by controlling the emission rate. Then there is the engine coolant temperature sensor, which monitors and signals temperature changes in the engine.

Various other types of sensors ensure driver and passenger safety. The anti-lock brake system measures wheel speeds, prevents skidding and allows steering control. Finally, new car models have ultrasonic parking sensors and rear facing cameras, which alert the driver with warning signals if **they** are going to collide with another vehicle or object.

Questions

- 1 Give the text a title.
- 2 What do the words in bold refer to?
- 3 Write synonyms for the following words.
 - a make sure
 - b crash into
 - c monitoring
 - d needs (n)

Part Two: Grammar and structure (60 marks)

II Complete the paragraph with words from the list.

vibration failure loose connecting
noises leaks disassemble site
internal

Maintenance and repair of home appliances are the responsibilities of repairmen. However, their job is not limited to repairing home appliances, they are also responsible for installing these appliances and sometimes (1) them to a gas or water pipeline. Once this is done, they need to check the pipelines in case there are (2)

If you have a problem with one of your home appliances, you ought to call a repairman and ask him to make a (3) visit in order to visually examine the appliance and detect its fault. The cause of (4) might be recognised by loud (5), immoderate pipe (6) or (7) parts. If one of these indications exists, the repairman might have to (8) the appliance and look for signs of corrosion in the appliance's (9) parts. If the fault isn't in these parts, then the technician has to examine the electrical connections.

III Complete the dialogue.

- Saber:** This is Syria Lift Instalment and Repair Company. Saber speaking. How can I help you?
- Customer:** My name is Noha Ramadani.
(1)
- Saber:** What seems to be the exact problem, Madam?
- Customer:** (2)
- Saber:** Is it insured?
- Customer:** (3)
- Saber:** (4)
- Customer:** Isn't today possible?
- Saber:** I'm afraid not.
- Customer:** (5)
- Saber:** In the meantime, stop using it to avoid further damage.

IV Change the following sentences into reported speech.

1 He asked, "Where will the meeting take place?"

.....

2 The authorities asked, "Why did the plane land at a different airport?"

.....

3 The teacher asked, "Who can draw this picture?"

.....

4 He asked, "Who is going to be nominated vice president?"

.....

V Choose the correct word in brackets.

1 I don't want to (do / make) the homework for this class.

2 Are you (doing / making) progress on the new building?

3 We (did / made) an agreement with the opposing party last week.

4 Did you (make / do) the laundry?

VI Write correct or incorrect next to each sentence. Change the incorrect ones to passive sentences.

1 The phone will answer automatically.

.....

2 They have been informed of this plan before.

.....

3 My essay must write today.

.....

4 The medication should stop if it begins to produce side effects.

.....

Part Three: Writing (25 marks)

IV Write a paragraph in which you describe the advantages and disadvantages of being a repairman.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

7 Inside the System

Exercise 1 Vocabulary

Match the underlined phrases or words in column A with their definitions in column B.

A	B
1 I just bought a new <u>laptop</u> to do my school work on.	a a flat panel that displays images
2 The <u>CPU</u> stopped working, and I had to take my computer to the technician.	b a guide to how computers work
3 To fix the CPU, the technician had to access the <u>motherboard</u> .	c the brain of the computer where most calculations take place
4 I tried to use a <u>computer manual</u> to solve the issue myself, but I was not successful.	d a computer device that is portable
5 Ahmed had to buy a new <u>screen</u> after he dropped his laptop and broke it.	e a circuit board that contains the main components of a computer

Exercise 2 Grammar

A Complete the sentences using the verbs in brackets.

- 1 If you inserted a video card into your computer, this (speed up) your computer.
- 2 It would be profitable and cheaper if you (buy) a computer from this brand.
- 3 If the computer (have) a larger storage capacity, it (perform) much better.

B Combine the given sentences using the second conditional and the proper form of the verbs.

- 1 I get a laptop as a gift. I download a lot of computer games.

.....

- 2 Something goes wrong with my computer. I call a computer technician for help.

.....

- 3 Your hard drive is damaged. All your data is lost.

.....

- 4 Her headphones don't work. She cannot listen to her favourite music.

.....

- 5 My computer's RAM is 4 Gb. It is much faster than before.

.....

- 6 Unlike you, I choose to buy a desktop computer, not a laptop.

.....

Exercise 3 Dialogue

Complete the dialogue with phrases from the box.

- That's right. The tasks that a graphic designer does are much more complex.
- You need a lot of storage space.
- I think so, too, because I mainly need the computer for Internet research and typing documents.
- Can you help me?
- For you, a 500 Gb hard drive is sufficient. But for me, I need a 250 Gb SSD.

Rami: Hello, Shadi. I want to buy a computer, but I'm not sure which one is the best for my job.
(1)

Shadi: Hello, Rami. I'm buying a computer, too! Let's do some research together. I think a 4 Gb RAM is enough for you as an editor.

Rami: (2)

Shadi: Yes, that's quite enough. For me, an 8 Gb RAM would be much better. You know, I need to download and edit photos, and the design software I use is quite advanced.

Rami: (3)

What about the hard drive? I'd love to have a fast computer. We always have very tight schedules.

Shadi: (4)

Rami: I agree. (5)

Exercise 4 Reading

Read the text and answer the questions.

Advancements in Computer Hardware

Fortunately, computer hardware has advanced considerably since it was first invented. The first computers were very large. Now, you can hold a computer in one hand. They were also very slow in the past. Now they are faster and computer manufacturers are constantly working to increase their speed. In other words, computer hardware innovation is constantly changing and improving.

The introduction of faster processors made computers more efficient. Also, video cards that possess their own processors and memory are now available, which is great for people who use computers for gaming. Hard drives are now smaller and are able to hold more information. The amount of available RAM has also increased significantly. Furthermore, it is now possible to have computers that are totally wireless. Interestingly, manufacturers are also focusing on producing 'green' computers. Major computer manufacturing companies are aware that they have a responsibility to protect the environment and to prevent a build-up of hazardous materials.

So what does the future hold for computer technology? There is the potential for further improvements. We can already unlock some computers biometrically rather than with a password. One day, we may not require a keyboard and may be able to use our computers just with hand gestures and eye movements. This type of technology is called 'perceptual computing' and it may be in your household in the not too distant future!

Questions

- 1 Which pieces of hardware mentioned in the text have been improved to help computers become more efficient?
- 2 Find words in the text which mean:
 - a new idea, method that is used for the first time.
 - in a noticeable or important way.
 - the duty to make sure that something is done well.
 - the possibility that something will develop or happen in a particular way.
- 3 Write an introductory sentence and a concluding sentence for the text.

Exercise 5 Writing

Use the notes to write in your own words an advert for a laptop. Lay it out like an advert in the box below.

**Notes:**

Size of screen: 14 inches

Battery life: 8 hours

CPU: 3 GHz

RAM: 8 Gb

Hard drive: 500 Gb

Price: \$500

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

8 Robots

Exercise 1 Vocabulary

Fill in the blanks with the correct words from the box.

popular rapidly software robotic humanoid

After breaking my vacuum cleaner, I decided to make my life easier and buy a (1) one. The robot has navigation (2), which helps it to detect the very dirty areas on the floor. This kind of robot has become very (3) recently, and you can already find it in a lot of homes. I can't wait for (4) robots to become available in the market. This way, I would have a helping hand at home. I am hoping for this kind of technology to be realised in the near future as the advances in technology have been developing very (5)

Exercise 2 Dialogue

Complete the dialogue with phrases from the box.

- How does Sally move her arms and legs like we do?
- Could you please tell us a little bit about her before you introduce her to us?
- So can we meet her now?
- What will Sally do?
- Is Sally an android?

Robotics engineer: Hello. Now before you meet my humanoid robot, make sure you are sitting quietly as she is very sensitive to noise.

Student: (1)

Robotics engineer: Yes, of course. Her name is Sally. She is very sociable and interactive.

Student: (2)

Robotics engineer: Sally has a head, face and body like a human. As she looks like a female human, she is called a gynoid. An android is the male humanoid robot.

Student: (3)

Robotics engineer: Sally has something called 'actuators'. These are motors that act like your muscles and joints and that allow her to move like you do.

Student: (4)

Robotics engineer: She will suggest games for you to play. Sally is made of different kinds of sensors that allow her to detect and hear you. And this is how she will be able to interact with you.

Student: (5)

Exercise 3 Grammar

A Complete the sentences using the correct form of the verbs in brackets.

- 1 If I (have) any idea about robots, I would not have asked you about it.
- 2 If you (share) your new ideas in robotics, many people would have been impressed.
- 3 If you hadn't looked after your robotic vacuum cleaner, it (break) down a long time ago.
- 4 If factory workers (use) masks, they would not have inhaled dangerous gasses.
- 5 If we (organise) more campaigns about the benefits of robots, people would have been more aware of them.
- 6 If factories had not had robots to help them, production of goods (be) very slow.

B Write six sentences of your own about robots using the third conditional.

.....

.....

.....

.....

.....

.....

Exercise 4 Reading

Read the text and answer the questions.

Assistance of Robots

Robots are very helpful machines. In manufacturing, they are able to produce better work than humans can, as they do not make mistakes and can work continuously faster and more precisely than humans. In addition, they can do dangerous jobs, for example working in unhealthy environments or in dark areas. Robots can also lift heavy objects. All this helps protect a human worker from injury. By using robots, companies can save a lot of money, as the robots save time and do not waste expensive materials through human error. Robots have also created more interesting jobs in programming for workers who would otherwise be doing the repetitive or dangerous jobs that are now done by robots. As robotics technology develops, robots will do more and more everyday jobs that we find boring and monotonous, allowing us to enjoy more free time.

Questions

- 1 Why is it better to have robots rather than humans in manufacturing?
- 2 How do robots save companies money?
- 3 What kind of jobs can humans do, now that robots are doing the repetitive and dangerous jobs?

Exercise 5 Speaking

Look at the picture of a home robot and a manufacturing robot. With a partner, carry out a conversation and discuss how these robots are similar and different depending on their work environment.



Exercise 6 Writing

Use the following sentences and phrases to write a short paragraph describing the job of a home robot.

- A home robot’s job includes ...
- Sometimes, depending on robots too much can be dangerous since ...
- If robots hadn’t been made ...

.....

.....

.....

.....

.....

.....

.....

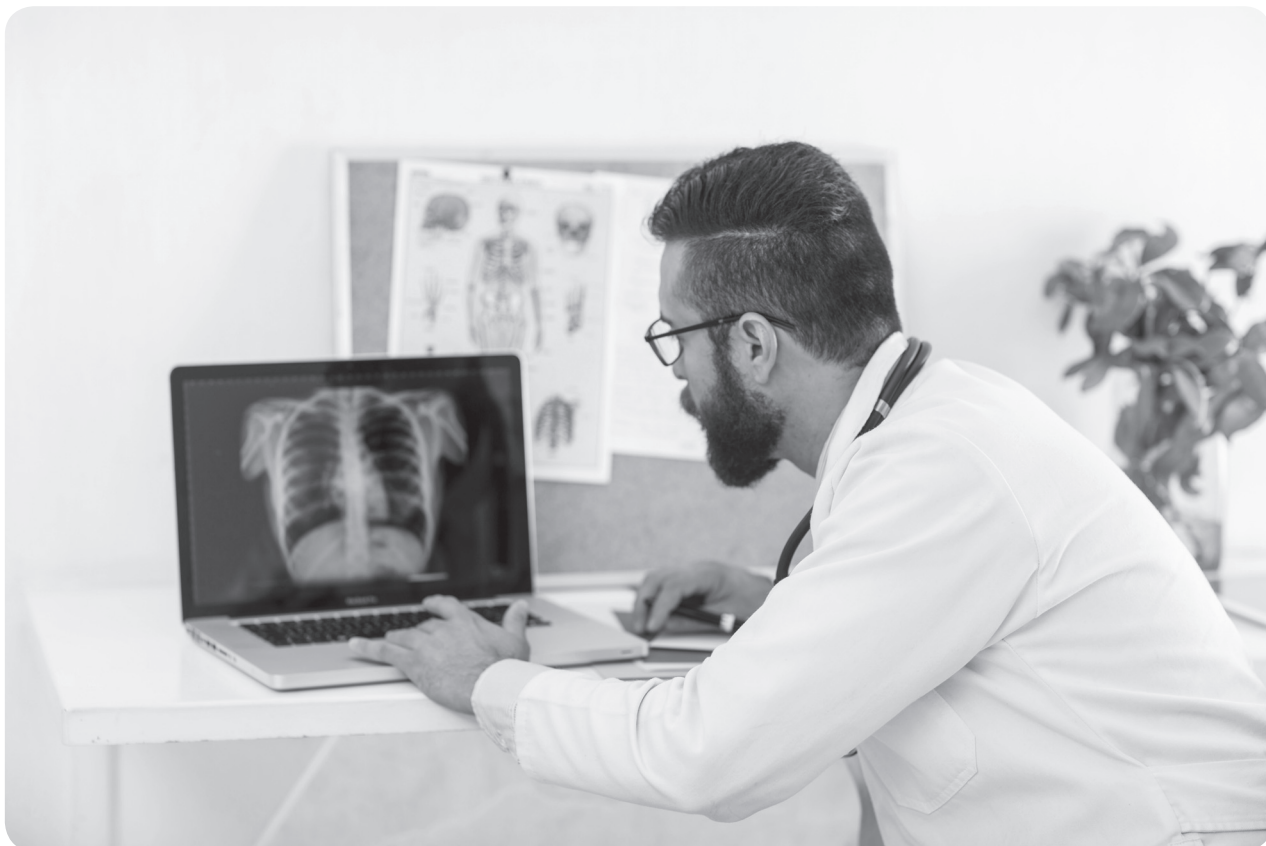
.....

9 X-rays

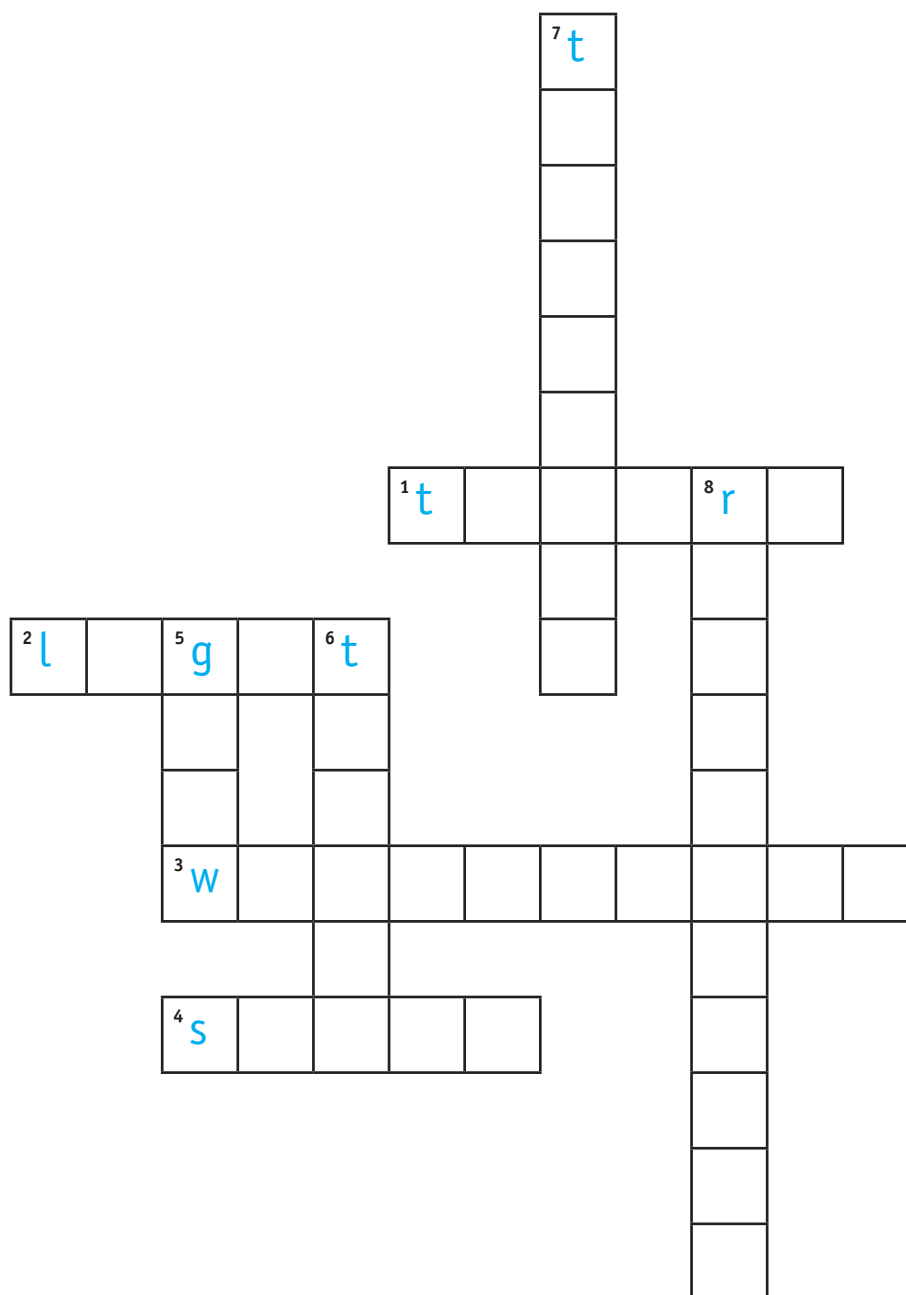
Exercise 1 Vocabulary

A Use the following words in a meaningful sentence.

- 1 theory
- 2 treatment
- 3 radiography
- 4 examination
- 5 tumour
- 6 experiment
- 7 discovery



B Complete the following crossword.



ACROSS	DOWN
1 an idea that is not yet proven to be true	5 to produce steady light
2 a natural source that makes things visible	6 to move a distance
3 the distance between different summits of a wave	7 medical care used to cure a certain illness
4 related to the sun	8 the process of taking a picture using X-rays

Exercise 2 Grammar

Complete the sentences below with the right tag question.

- 1 X-ray astronomy is a modern way of observing astronomical objects, ?
- 2 X-ray machines don't need any maintenance, ?
- 3 She has received an X-ray therapy, ?
- 4 The doctor did not call anyone to repair the X-ray machine, ?
- 5 The employees can have the office ready by Monday, ?
- 6 X-rays are indispensable nowadays, ?

Exercise 3 Reading

Read the text and answer the questions.

.....

The invention of the X-ray machine is one of the most important achievements in medical history. Without it, life would be so much harder; broken bones or serious illnesses could not be detected. This means that a lot of injuries and diseases would go undetected and more serious problems would occur. Luckily nowadays, X-ray machines are widely available in hospitals. They can be basic X-ray machines, or the more advanced CT scanners. It is important to note that there are other machines that are able to take images of our bodies that do not use X-rays. These use nuclear medicine and are called PET scans. There are also ultrasound and MRI machines. The doctor will decide which type of machine is needed, depending on the patient's symptoms.

Questions

- 1 Give the text a suitable title.
- 2 According to the text, what problems would occur if X-rays did not exist? Mention other problems.
- 3 Are X-rays the only machines that can take images of our bodies? Explain your answer and give examples.
- 4 Find words in the text which mean:
 - something done successfully, typically by effort, courage or skill.
 - using the most modern ideas, equipment and methods.
 - the production of something for the first time.

Exercise 4 Writing

Write a short paragraph stating two advantages and two disadvantages of X-ray therapy for cancer patients. You may use some of the information provided in the fact box below to help support your ideas.

X-ray Therapy Fact Box

Radiation therapy uses X-rays to kill cancer cells or shrink a tumour.
X-ray therapy can cure patients from cancer completely, or it can slow down the growth of the cancer cells.

Test 3

Part One: Reading comprehension (35 marks)

I Read the following text and answer the questions.

Have you ever thought about how we used to cope without computers? Have you ever considered their effect on how we live today? It's interesting to know when and how computer hardware came into existence and to take a look at the first basic machines.

The earliest tool used to help humans with digital calculations was a simple tally stick. The tally stick was an animal bone or piece of wood that was used to carve numbers, quantities and messages on. It dates back to the Upper Paleolithic period – that's between 50,000 and 10,000 years ago! Later, ancient civilisations used the abacus for arithmetic tasks. The abacus is a simple frame with beads sliding on a wire, which is still used today to teach children. Originally, beans or stones would have been used instead of beads. A more sophisticated abacus known as "Napier's Bones" was designed in 1617 by a Scottish mathematician and physicist and was a useful aid for multiplication. Later during the 17th century, the first mechanical calculator was created in Germany. In 1820, a French inventor created the first mass-produced mechanical calculator.

An English engineer called Charles Babbage is considered to be the 'father of the computer'. In the 19th century, he invented the first ever mechanical computer. Unfortunately, due to lack of funds, Babbage had to abandon his project. However, his son Henry Babbage managed to make a more simplified version of his father's creation. Later in the 20th century, analogue computers were invented. These computers used a mechanical or electrical model of a problem for computation. Whilst they were sophisticated at the time, they were not programmable or very accurate. The principle of the modern computer was due to work by Alan Turing, who designed the universal Turing machine.

Today, technological advancements have led to computers being everywhere. Let's stop and ask ourselves: could we live without them?

Questions

- 1 What was the name of the first tool used for recording data?
- 2 Which tool was used by ancient civilisations and is still used today?
- 3 Who is known as 'the father of the computer'?
- 4 What was the problem with analogue computers?
- 5 Name two places where computers are essential.
- 6 Do you think we could ever go back to a time without computers?
- 7 Suggest a title for the text.

Part Two: Grammar and structure (60 marks)

II Complete the paragraph with the correct form of the words in brackets.

All businesses use computer technology for (communicate) (1) between their employees. Computers are an (essence) (2) means of (storage) (3) information, data, images and accounts. They allow people to share information (quick) (4) Secretaries will plan the agenda of the manager using a program that allows the manager to see the (day) (5) schedule. The accounting department have programs to (accurate) (6) monitor company finances. All (employ) (7) are able to send emails to each other and to clients. During their breaks, they use their computers for (person) (8) interests like searching the Internet for holidays or watching videos.

10 Automatic Control

Exercise 1 Vocabulary

Circle the odd word out.

- | | | | |
|---------------|-------------------|--------------|---------------|
| 1 a computer | b printer | c dishwasher | d grater |
| 2 a automatic | b manual | c reaction | d independent |
| 3 a function | b use as | c act as | d formula |
| 4 a mistake | b maintenance | c correction | d adjustment |
| 5 a device | b characteristics | c object | d tool |

Exercise 2 Word formation

Put the words in brackets in their correct form.

- 1 A computer is an example of an (automate) control system.
- 2 The systems studied within automatic control are mostly the (line) systems.
- 3 In automatic control systems, a controlling device regulates the (operate) of a machine with (minimum) human intervention.
- 4 The First and Second World Wars saw major (advance) in the field of automatic control systems.



Exercise 3 Dialogue

Complete the following dialogue.

A: What would I use an automatic control system for?

B: It can be used to help simplify your life.

A: How can it do that?

B:

A: Give me an example from our daily life.

B:

A: So, in other words, what you're saying is

B:



Exercise 4 Grammar

A Complete the text using an indefinite pronoun from the box.

both one all everybody somewhere
anything any everyone

(1) innovations have brought some change in (2) 's lives.
(3) old and recent innovations derive from the need to make life easier and more practical. (4) of the most important innovations in history is the aeroplane, which can transport people and goods to any country in the world. If you choose to go (5) outside your country, catching an aeroplane would be the easiest and fastest means of transportation available to (6) You can also ship almost (7) to (8) country in the world, and it would arrive in a relatively short time. Aeroplanes have been a crucial innovation in the history of transportation.

B Write six sentences of your own about automatic control using indefinite pronouns.

.....

.....

.....

.....

.....

.....

Exercise 5 Reading

Read the following text about the function of automatic control systems and answer the questions.

Control systems have been developed to ease humans' lives. Many tasks that people need to do or oversee are too difficult to control. As a result, control systems make it easier by carrying out the tasks that humans find difficult to do in a shorter period of time. At the same time, automatic control systems save money and reduce risks. The basic functions that these systems carry out include measurement, comparison, computation and correction. Such functions are controlled by measurement, error detection and final control elements. They all work together to help take over a task humans find difficult.

Questions

- 1 What is the main idea of the text?
- 2 What functions do automatic control systems carry out?
- 3 Find the meaning of the following words in the dictionary:
 - a variable
 - b oversee
 - c hazardous
 - d error



Exercise 6 Writing

Write a short paragraph discussing the importance of automatic control systems, based on what you have learned in this lesson.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



11 Electrical Power Stations

Exercise 1 Vocabulary

Reorder the jumbled letters to find the term that fits the definition.

- 1 ahurldyic operated by a liquid
- 2 rolas relating to the sun
- 3 rarnegote a piece of equipment that converts energy into electricity
- 4 bintrue a machine for producing power by a wheel or rotor
- 5 celercyitit a form of energy resulting from charged particles

Exercise 2 Grammar

A Choose the correct preposition from the box.

by on in across for
of to at before

- 1 Our science teacher took us a field trip the power station.
- 2 A power station provides electricity buildings.
- 3 Solar power and wind power are examples energy sources.
- 4 Were you talking about electric power stations I came?
- 5 We will be looking power station pictures.
- 6 The first power station started operating 1882.
- 7 This was the first electric tramway to be powered hydroelectricity.
- 8 Electric pylons for carrying cables are being built navigable rivers.

B Read the text below. There are FIVE mistakes in the use of prepositions. Correct them by replacing them with the appropriate preposition from the box.

into on after by(x2)

Our school took us to a trip to the city's power station. We went on bus, since the distance was too far to walk. There, we saw how electricity was produced and then distributed around the city. We went in one of the control rooms and observed the technicians at work. Before we saw the control room, we then went to learn how the generators of the station worked. The turbines were moved on water or wind currents.

Exercise 3 Reading

Read the text and answer the questions.

Power stations generate electrical power. Many power stations use thermal (heat) power to create energy. However, there are other sources of energy used to generate electricity, such as water, sun and wind. These three sources are transformed in hydroelectric, solar and wind plants respectively. Hydroelectricity is produced through dams built as reservoirs that store water. This water runs through and moves turbines which are connected to electricity generators. In solar power plants, photovoltaic cells, or solar panels, turn sunlight into electricity. Wind turbines also create electricity in wind power plants. It is vital that such plants are located in areas with strong, steady winds. Electricity can therefore be generated in more than one way.

Questions

- 1 Give the text a suitable title.
- 2 What are the four sources of energy that could be used to generate electricity? Briefly describe each.
- 3 What is an important condition for a wind power station to be set up and why?
- 4 Define the following words, using context clues from the text:
 - a thermal
 - b power station
 - c solar
 - d dams



Exercise 4 Writing

Think about the location where you live. Consider what sources of energy you have most in that area: thermal, fuel, water, wind or sun. Write a short paragraph discussing what type of power station you think would be most suitable to produce the most energy in your area.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



12 Latest Discoveries

Exercise 1 Vocabulary

Write the words in brackets in the right form.

Dragging your luggage around with you is a (pain) **(1)** task, especially on the way back from a holiday. This new (invent) **(2)** allows you to stop pulling around your luggage or wheeling it back home. This luggage bag is a mini scooter, allowing you to scoot around with your luggage (safe) **(3)** tucked away. When everyone else is carrying around their luggage, you will be jumping on your scooter, shortening your trip back home and making the journey more fun. The scooter also folds up, allowing you to use it as a normal luggage carrier. It's a simple idea but an (extreme) **(4)** effective product, which we think will become very popular.

This gadget allows you to recharge four devices (simultaneous) **(5)** It is compatible with over 1,000 products. It is a (practice) **(6)** and (afford) **(7)** charging station and it is offered in white or black. It is an extraordinary good looking product, designed for the office and every room in the house. Its advantage is that you can (charge) **(8)** most of your electronic devices simultaneously.

Exercise 2 Dialogue

Imagine you are a salesperson standing on the road trying to convince passing customers to buy the following item. Write a dialogue between you and one of the customers.

This new LED umbrella comes equipped with a built-in light which illuminates in the dark. At the touch of a button, the light shines offering you a safe and secure passage to your destination. The light is embedded inside the handle to lighten the area under the umbrella's hood. For those of you who can't handle a long walk home in the dark, this LED umbrella is sure to make that walk a little brighter.



Exercise 3 Grammar

A State whether the following sentences are fragments or run-ons.

- a While many people have digital cameras these days.
- b Though advancements in microelectronics made the vision of a computer a realistic one.
- c All televisions make use of an image which is scanned to produce a representation time signal, it is then reversed and decoded by the human eye.
- d It's hard to decide which inventions are the most important, they are so many.
- e The transistor possibly the most important of the famous inventions of the twentieth century.
- f Received a patent that was needed to bring about the industrial age.
- g The engineer whose high-frequency alternator gave America its start in the field of radio communication.
- h The Internet something which many of us now take for granted.
- i Through various experimentations with electricity, he invented a basic electric battery, it was followed by electric light once he realised that carbon glowed, producing light when connected to the battery.
- j These great men and their ideas have changed the world, the way we live our lives.

B Correct the fragments and run-ons in task A.

Exercise 4 Taxonomic skills

Unscramble the following sentences to make two advertisements about two new inventions.

- a The blues and greens indicate a mellow mood, compared to pinks, reds and oranges which indicate a stressed-out individual.
- b The stress watch is aimed at busy people, with hectic lives, as it monitors stress levels throughout the day and indicates your levels via a black bar graph on the screen.
- c There is no better alarm system than the round of a vicious dog barking ferociously at the intruder's awaiting entry point.
- d Once the stress levels hit a peak, you can take a rest and watch as your levels begin to drop.
- e The alarm noise is sounded whenever the door knob is touched or door is pushed, so the alarm is designed to be set off under the slightest of pressures.
- f This invention is called an Alarm Security Bar which adds instant protection to any home without the need for expensive installation and monitoring.
- g Thankfully, the watch also tells the time, so you will never miss an appointment or a meeting.
- h This watch does not only look fantastic, but also serves a great purpose.
- i The alarm can either be set to a powerful siren or the alternative barking alarm sound.



Exercise 5 Reading

Read the text and answer the questions.

Can you believe that going solar is cheaper than burning coal? Nanosolar is a new solar power invention without a panel but just a thin coat of paint that takes light and converts it to electricity. It draws power from the air; it is popular because it is cheap.

The company produces its power sheet solar cells with printing-press-style machines that lay down a layer of solar-absorbing nanoink onto metal sheets as thin as aluminium foil, so the panels can be made for about a tenth of what current panels cost and at a rate of several hundred feet per minute.

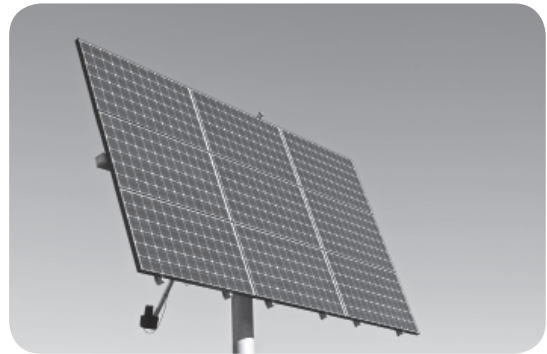
Cost has always been the biggest problem of solar panels. Traditional solar cells require silicon, and silicon is an expensive commodity. It also has to be put on glass, so it's heavy, dangerous and expensive to ship and install. Nanosolar cells use no silicon.

Right now, the biggest question for Nanosolar is not if its products can work, but rather if it can make enough of them. California, for instance, has recently launched the Million Solar Roofs initiative, which encourages the installation of 100,000 solar roofs per year, every year, for 10 consecutive years.

Questions

- 1 Give the text a suitable title.
- 2 Match the words with their meanings.

a convert	1 existing
b current	2 new idea or project
c launch	3 uninterrupted
d initiative	4 change
e consecutive	5 begin
- 3 List two advantages of this new invention.
- 4 What are the disadvantages of the old solar systems?
- 5 Do you think this new invention is going to be implemented in many countries in the world? Why? Explain.



Exercise 6 Writing

Can you imagine an electronic glove that allows you to write in thin air? It looks very similar to a fingerless bike glove, but it has a built-in sensory computer program. It can recognise when you trace letters of the alphabet. What are the advantages and disadvantages of such a pen? Write a well-organised paragraph.

.....

.....

.....

.....

Test 4

Part One: Reading comprehension (35 marks)

I Read the following and answer the questions.

Here are stories of two interesting inventions. The first is that of the English inventor and businessman Percy Shaw (1890–1976). Shaw lived in London where the streets were very dirty and the air was polluted with smog- a mixture of smoke and fog. That made it difficult for people to see the roads. Shaw wanted to find a means to light the roads. One evening, while he was driving, he noticed that his car lights reflected the eyes of the cats on the streets. This is how he invented the 'Catseyes' we see now in the middle of the road to guide people to keep in lane.

The second invention is that of the American actress Mary Anderson (1859–1940). During a trip to New York in 1903, she noticed that drivers put their hands out of the vehicle they were driving to clear the windscreen. Anderson wanted to find a way to help drivers clean rain, snow or dirt from the windscreen and not get wet. The following year, she applied for a patent for a device that consisted of a lever that a driver could operate from inside the car. The lever caused a spring-loaded arm with a rubber blade to swing across the windshield and then back again to their original position. Anderson got the patent for the device in 1905.

1 Find a word in the text that means:

- a** one of a line of small flat objects fixed in the middle of the road that shine when lit by car lights and guide traffic in the dark
- b** the legal right to be the only producer or seller of an invention
- c** the large window at the front of a vehicle (a car, bus, etc.)

2 Did Shaw and Anderson intend to be inventors, or did they become inventors by chance? Explain your answer.

.....
.....
.....
.....

3 The word 'smog' is derived from two words: smoke and fog. This is called the process of blending. Write down the two words used in each of the following:

- a** sitcom
- b** brunch
- c** camcorder

4 Do you think anyone can invent a useful device? Could you?

.....
.....
.....
.....

II Complete the dialogue. Use the appropriate punctuation.

- you are consuming.
- I wonder what new inventions we'll have soon.
- Can you guess what it is?
- How often do you check what it's recording?
- I'll tell you.
- Where do you keep it?

A I've just bought this device from the shop.

(1)

B Well, it seems like an electronic device to measure something.

A Yes, that is close. (2)
It's a digital machine that keeps record of your heartbeats and beeps when your heartbeats are irregular.

B Oh, how interesting! It reminds me of another medical device that measures how many calories (3)

A (4)

B It has a strap that you can fasten around your wrist or ankle.

A (5)

B You have to check it every twelve hours.

A (6)

III Choose the correct word in brackets.

- 1 I met (somebody / anybody) on the bus whom I knew at school.
- 2 They didn't buy (something / anything) at the exhibition of new discoveries.
- 3 Samia found (nothing / nowhere) that she wanted to decorate her new dress with.
- 4 John couldn't find his suitcase (anywhere / somewhere)
- 5 I assured them they would find the product they're looking for (nowhere / somewhere)

IV Circle the correct preposition.

- 1 Mum put the blanket (up, on, under) the bed when it became cold.
- 2 Mona found her notes (along, through, between) her books.
- 3 The child hid his new toy (over, under, along) the chair.

- 4 He came (along, up, with) us to the power station.
- 5 The chef has put the two trays (into, between, through) the oven.
- 6 The guard (into, at, up) the exhibition was attentive.

V Correct the following sentences.

- 1** The advertisement one of the best I've ever seen.

2 They produced the new kitchenware put it in the market they did not advertise it.

- 3 When I read the newspaper.

- 4 If they buy the car.

- 5** Since I saw Mona last.

Part Three: Writing (25 marks)

VI Choose a profession and an appropriate place of work. Write a brochure that gives a list of safety instructions for people at work.

Glossary

A

absorb *verb* to take in or soak up energy, a liquid or other substance by chemical or physical action, typically gradually: *Buildings can be designed to **absorb** and retain heat.*

accurate *adjective* (of information, measurements, statistics, etc.) correct in all details; exact: *It is important that the doctors give their patients **accurate** information about their illness.*

adjust *verb* to alter or move something slightly in order to achieve the desired fit, appearance or result: *The interest rate should be **adjusted** for inflation.*

advancement *noun* a development or an improvement: *The **advancements** of technology have helped humanity move forwards.*

affordable *adjective* inexpensive: *The prices of this shop are **affordable**.*

allocate *verb* to officially decide that a particular thing or amount of money, time, etc. should be used for a particular purpose: *The charity has been **allocated** a big sum of money through donations.*

alternative *adjective* different from what is usual, expected or traditional: *Solar energy is an example of **alternative** energy.*

android *noun* a robot that looks completely human: *The villain in this film is an **android**.*

angiogram *noun* an X-ray photo of blood or lymph vessels: *The **angiogram** showed that he has a heart disease.*

annual *adjective* occurring once every year: *I always attend the union's **annual** conference.*

appliance *noun* a device or piece of equipment designed to perform a specific task, typically a domestic one: *He sells imported home **appliances**.*

apprenticeship *noun* the process of learning or training under a professional or skilled worker: *I learned everything about the mechanics of a car because of my **apprenticeship**.*

assemble *verb* to fit the pieces or parts together: *At the factory, we **assemble** used car parts to make new cars.*

automatic *adjective* works by machine: *The **automatic** door opens by itself.*

automation *noun* the use of control systems and information technologies to reduce the need for human work in the production of goods and services: *Unemployment has increased due to the spread of **automation**.*

awful *adjective* used to emphasise the extent of something, esp. something unpleasant or negative: *The fridge smells **awful**.*

B

back up *verb* to reserve or make a copy to store: **Back up** your files before you defragment your computer so that nothing is lost.

biometrically *adverb* using biometric methods; referring to the detailed information in someone's body like the colour of their eyes, etc.: DNA is measured **biometrically**.

blower brush *noun* a tool useful for cleaning a microscope, telescope or camera and can be used with fingerprint dusting powder for science projects: The photographer used a **blower brush** to clean the lens of his camera.

boiler *noun* a container for boiling water that is part of a steam engine or is used to provide heating in a house: This family has no access to hot water at their house because their **boiler** is broken.

boot *noun* an enclosed space at the back of a car, used for carrying bags: They placed their luggage in the **boot** of the car.

bulky *adjective* something that is bigger than other things of its type and is difficult to carry or store: It was difficult for him to carry the box because it was very **bulky**.

bump *noun* a small raised area on a surface: The moving arm on record players turns the **bumps** in the plastic into sounds.

bumper *noun* a bar fixed on the front and back of a car to protect it if it hits anything: My car's **bumper** fell off during an accident.

burglar *noun* someone who steals things: A **burglar** broke into a family's house.

C

cabin *noun* the area for passengers on a lift: The **cabin** could only fit five people, so some of us had to take the stairs.

cable *noun* strong, heavy and thick rope or wire: The **cable** used to carry a lift is made of very tough material.

cancer cell *noun* a cell that constantly divides, forming solid tumours or flooding the blood with abnormal cells: After getting tested, she found out that **cancer cells** were present in her blood.

carer *noun* someone who looks after an old or ill person at home: **Carers** sometimes work for free.

carry out *phrasal verb* to conduct; to perform: They **carried out** the operations in secret.

casing *noun* a cover that protects something: Protective **casing** is needed so that you make sure you don't damage your laptop.

CD-R *noun* a blank compact disc that can be recorded on only once: She added her favourite songs to a **CD-R**.

CD-ROM *noun* a CD on which large quantities of information can be stored to be used by a computer: He inserted his **CD-ROM** in the CD player.

CD-RW *noun* a blank CD that can be recorded, erased and re-recorded many times: He erased the data on his **CD-RW** to replace it with newer data.

chemical *adjective* connected with or used in chemistry, or made by a chemical process: The students were witnessing a **chemical** reaction in the science lab.

clutch *noun* the pedal that you press with your foot when driving a vehicle in order to change gear, or the part of the vehicle that this controls: Put your foot down on the **clutch** before you change gear.

coil *noun* circular wire: *The coil in the light bulb heats up to give light.*

communications engineering *noun* the science of ways to connect with people and places (telephone, internet, etc.): **Communications engineering** has helped people to talk to each other even though they are very far apart.

compartment *noun* a smaller enclosed space inside something larger: *The refrigerator is divided into different compartments, each for a specific use.*

condenser *noun* a machine that puts pressure and turns a gas into a liquid: *The water vapour passes through the condenser and comes out as water.*

conference *noun* a formal meeting for discussion: *He gathered all the men around the table for a conference.*

considerably *adverb* much more: *Her health was getting considerably better.*

control panel *noun* an electrical device that has switches that control another device: *Samir is able to control the whole machine from his office, with a control panel.*

control system *noun* is a device or set of devices to manage, command, direct or regulate the behaviour of other devices or systems: *The professor said that my thesis has missing information about applications of control systems.*

conventional *adjective* standard; traditional: *People stopped buying conventional cameras ever since the invention of the digital camera.*

convert *verb* to cause to change in form, character or function: *The procedures followed converted raw material into useful forms.*

counterweight *noun* a weight that, by exerting an opposite force, provides balance and stability of a mechanical system: **Counterweights** are used in lifts.

CPU *noun* Central Processing Unit; the part of a computer that controls and organises all its activities: *A CPU can perform basic mathematical operations.*

crew *noun* a group of people working together with special skills: *They are the most efficient TV camera crew.*

CT scan *noun* a special X-ray test that produces images of the body using X-rays and a computer: *A CT scan is often used to examine people who could have internal injuries.*

cure *noun* a medicine or medical treatment that can cure an illness or condition: *This disease has no cure.*

D

defragment *verb* to reorganise a computer file: *Tarek defragmented his computer to make it run faster.*

delicate *adjective* very fine in texture or structure: *The sun can easily damage a child's delicate skin.*

dense *adjective* having a lot of mass: *This land is dense with vegetation.*

diagnostic *adjective* concerned with the diagnosis of illness or other problems: *An MRI is a type of medical diagnostic test.*

diode *noun* a semiconductor device with two terminals, typically allowing the flow of current in one direction only: *Avalanche diodes can protect circuits from a high voltage flow.*

downside *noun* the negative side of something: *Lack of sleep has many **downsides** like stress and fatigue.*

dual *adjective* having two of something or two parts: *This book has the **dual** role of informing and entertaining.*

DVD-R *noun* a digital versatile disc that can be recorded on only once: *She transferred videos of her childhood from an old video cassette to a **DVD-R**.*

DVD-RAM *noun* a DVD that can be repeatedly read, written on and erased: *He deleted old videos from the **DVD-RAM** because he no longer needed them.*

DVD-ROM *noun* a read-only DVD commonly used for storing large software applications: *He inserted his **DVD-ROM** in the drive to play the new game he had bought.*

E

echolocation *noun* the location of objects by reflecting sounds: *Bats use **echolocation** to move around in the dark.*

effective *adjective* successful in producing a desired or intended result: *Her teaching method was very **effective** on the students' results.*

efficient *adjective* if someone or something is efficient, they work well without wasting time, money or energy: *My wife is an **efficient** secretary.*

electricity *noun* a source of energy: *We use **electricity** for light and heat.*

electrode *noun* a solid that allows an electric current to run through it: *Use an **electrode** to jump start a car if there are no cables available.*

equipment *noun* the tools, machines, clothes, etc. that you need to do a particular job, or activity: *They bought some brand new computer **equipment**.*

error *noun* mistake: *The **errors** in the machine were fixed by the technician.*

essential *adjective* absolutely necessary: *Water is **essential** for survival.*

Ethernet *noun* a system for connecting a number of computer systems to form a local area network: ***Ethernet** is the most used local area network technology.*

examination *noun* a detailed inspection or investigation: *She was nervous about her **examination** at the hospital.*

exceptional *adjective* unusual; not typical: *Tonight's meal was **exceptional**.*

exposure *noun* the state of being in contact with something: *Long **exposure** to the sun is harmful to the skin.*

Extraction-Assist Robot *noun* a humanoid robot designed as a method for transporting wounded soldiers from the battlefield to safety: *The battlefield **Extraction-Assist Robot** has the size and shape of an adult male human.*

F

face *verb* to confront and deal with something or a situation: *You'll just have to **face** the facts.*

facilitate *verb* to make an action or process easy or easier: *Our research has been **facilitated** by private funding.*

facilitator *noun* someone who helps a group of people discuss things with each other or do something effectively: *A **facilitator** is expected to offer guidance.*

fever *noun* an illness or a medical condition in which you have a very high temperature: *He did not go to school because he had a very high **fever**.*

fine *noun* a sum of money paid by someone who has broken the law: *He had to pay a **fine** for crossing the speed limit.*

fluoroscopy *noun* an imaging technique that uses X-rays to obtain real-time moving images of the interior of an object: *Before the discovery of **fluoroscopy**, doctors had to surgically open patients to observe their organs.*

forum *noun* a place, meeting or medium where ideas and views on a particular issue can be exchanged: ***Forums** were held for doctors to exchange their views on medical research.*

fossil fuel *noun* a fuel such as coal or oil that is produced by the very gradual decaying of animals or plants over millions of years: *Environmentalists would like to see **fossil fuels** replaced by renewable energy sources.*

G

generate *verb* to produce: *Some countries use water to **generate** electricity.*

generator *noun* a machine used to create energy: *The **generator** produces energy that is used to make electricity.*

green *adjective* not harmful to the environment: *This supermarket has banned the use of plastic bags and introduced **green** shopping bags instead.*

greenhouse gas *noun* a gas in an atmosphere that absorbs infrared radiation: *Methane and carbon dioxide are examples of **greenhouse gases**.*

H

hair-washing machine *noun* a machine for washing a person's hair or means to assist in the washing of it: *Many salons are using automatic **hair-washing machines** which include an outer as well as an inner shell.*

hard drive *noun* a non-removable magnetic disk with a large data storage capacity: *The computer's **hard drive** is not functioning well.*

hardware *noun* machines, wiring and other typical components of a computer or other electronic system: *The CPU and hard drive are two components of a computer's **hardware**.*

HDMI *noun* a standard for connecting high-definition video devices: ***HDMI** cables are mostly used for connecting game consoles with the television.*

heat *verb* to make warm or hot: *You need to **heat** the food in a microwave.*

hoist *noun* a piece of equipment used for lifting heavy objects with ropes: *The fisherman used a **hoist** to lift the big fish out of the water.*

humanoid *adjective* having an appearance or a character resembling that of a human: ***Humanoid** robots look so much like humans.*

humidity *noun* a quantity representing the amount of water vapour in the atmosphere or a gas: *The **humidity** level is very high today.*

hydraulic *adjective* involving or operated by a fluid: *The engineer checked the aeroplane's **hydraulic** systems.*

hydro *prefix* related to water: ***Hydroelectricity** is generated by flowing water.*

I

impressive *adjective* appearing very good, large, important, etc. so that you can admire it: *His grades this semester were **impressive**.*

industry *noun* a group of factories that produce goods: *The timber **industry** produces wood.*

innovation *noun* a new method, idea, product, etc.: *This technological **innovation** will be very useful in the medical field.*

inspect *verb* to study something carefully for any flaws: *The manager **inspected** the work of the new trainee.*

install *verb* to place or fix an equipment or machinery in position ready for use: *We're planning to **install** a new shower.*

internship *noun* a job that lasts for a short time and that someone, especially a student, does in order to gain experience: *She is looking for an **internship** at a law company.*

intersection *noun* a point at which two or more things intersect: *There was a car accident on the road **intersection**.*

invention *noun* the act of designing something new: *The **invention** of the control system has made life much easier.*

J

jack *noun* a device used to lift heavy objects: *He used a **jack** to help change the tyres.*

L

lens *noun* a piece of curved glass or plastic which makes things look bigger or smaller: *The **lenses** of her glasses need to be changed.*

lifespan *noun* the length of time for which a person or animal lives or a thing functions: *This battery has a short **lifespan**.*

lift *noun* a device used to carry passengers up and down a building: *We took the **lift** up to the 4th floor.*

lightweight *adjective* of thin material and weighing less than average: *This new phone is very **lightweight** and easy to carry.*

load *noun* heavy weight: *It increased the **load** on the wheels.*

M

magnetic field *noun* a vector field that describes the magnetic influence of electrical currents and magnetised materials: *Earth's **magnetic field** is believed to be generated in the Earth's core.*

maintain *verb* to keep in good condition: *You should constantly **maintain** your appliances.*

maintenance *noun* the repairs, painting etc that are necessary to keep something in good condition: *My dad asked the mechanic to include the cost of repairs and **maintenance** in the bill.*

manta ray *noun* a very large ray of warm seas that has fins that resemble wings: ***Manta rays** can't be sold, so they are thrown away if they were ever caught.*

manual *adjective* works by hand: *We used to use a **manual** pump to get water from the well.*

manual *noun* a book that gives instructions about how to do something, esp. how to use a machine: *Consult the computer **manual** if you have a problem.*

manufacture *verb* to make something on a large scale using machinery: *He owns a company that **manufactures** paint-by-number sets.*

material *noun* resources: *The **material** used to build a house includes wood and metal.*

mechanic *noun* one who repairs machines, cars or tools: *I took my car to the **mechanic** for a checkup.*

mechanical *adjective* working or produced by machines and machinery: *The air conditioning is a type of **mechanical** device.*

mechanism *noun* a natural or established process by which something takes place or is brought about: *We have no **mechanism** in place for assessing the success of forwarded inquiries.*

microchip *noun* a tiny piece of semiconducting material used to make an integrated circuit: *The veterinarian was able to identify the lost cat by its **microchip**.*

mill *noun* a building fitted with manufacturing machinery: *I would like to work in a steel **mill**.*

minimise *verb* to reduce: *Machines **minimise** the time it takes to do a job.*

modify *verb* to change: *I **modified** the measurements of the board to fit through the door.*

moisture *noun* small amounts of water that are present in the air, in a substance or on a surface: *Plants use their roots to absorb **moisture** from the soil.*

monotonous *adjective* lacking in variety and interest: *The speaker's voice was very **monotonous**.*

motherboard *noun* a printed circuit board containing the main components of a microcomputer or other device: ***Motherboards** used to have very few components, but nowadays, they have many built-in features.*

motor *noun* a machine or an engine that helps an object move: *If the **motor** of the car breaks down, the car would stop running.*

motorised *adjective* having an engine (used, esp. when something does not usually have an engine): *She has a **motorised** wheelchair.*

N

nickel *noun* a hard silver-white metal that is often combined with other metals, for example, to make steel: ***Nickel** is considered to be resistant to corrosion.*

notch *noun* a V-shaped cut or hole in a surface or edge: *An arrow has a **notch** on its end.*

O

operate *verb* to control the function of a machine: *The technician **operates** the machine.*

optical *adjective* of or relating to sight, especially in relation to the physical action of light: ***Optical** illusions use colour, light and patterns to create images that can trick our brain.*

optimal *adjective* best or most favourable: *This design makes **optimal** use of the available space.*

out of service *expression* not available for use: *"**Out of Service**" tags should be used to prevent the accidental operation of machines that are being maintained or cleaned.*

overhead *adjective* situated above the level of the head: *A plane flew **overhead**.*

overload *verb* to put too many things or people on or into something: *Be careful not to **overload** the washing machine.*

oversee *verb* to look over; to observe: *The manager **oversees** the work of his employees.*

P

Paper-Based Lithium-Ion

Battery *noun* an ultra-thin battery that could be used to power electronic newspapers:

***Paper-Based Lithium-Ion Batteries** were developed by Stanford researchers.*

peripherals *noun* extra or things not related to the main or central object: *The **peripherals** of the computer include the mouse and the speakers.*

Perpetual Motion Device

***noun** any device or system that indefinitely produces more energy than it consumes, resulting in a net output of energy for indefinite time: *The laws of physics have to be broken for the **perpetual motion device** to work properly.**

photosensitive *adjective* reacting to light, for example by changing colour or producing an electrical current: *Some television channels give a warning to people who have **photosensitive** epilepsy before they broadcast a programme with lots of flashing images and colours.*

pinpoint *verb* to find or show the exact position of something: *He was not able to **pinpoint** the exact location of where he really was.*

pixel *noun* an extremely small area of illumination on a display screen, one of many from which an image is composed: *Larger **pixels** usually offer a higher image quality.*

pole *noun* a long, cylindrical, often slender piece of wood, metal, etc.: *The truck backed into a metal **pole** and left it bent.*

polish *verb* to make something smooth, bright and shiny by rubbing it: *The man was **polishing** his black shoes.*

port *noun* a socket in a computer or network into which a device can be plugged: *He plugged his USB into the computer **port**.*

portable *adjective* able to be easily carried or moved, especially because of being a lighter and smaller version than usual: ***Portable** CD players were very popular in the 1990s.*

potential *noun* latent qualities or abilities that may be developed and lead to future success or usefulness: *This candidate has a promising **potential**.*

power supply *noun* a device that supplies power to at least one electric load: ***Power supplies** can convert different types of energy, like solar energy, into electrical energy.*

press *verb* **1** to make clothes smooth using a hot iron: *I'll need to **press** my suit.* **2** to push a button, switch, etc. to make a machine start, a bell ring etc: ***Press** the power button to turn on the computer.*

procedure *noun* an established or official way of doing something: *The **procedure** to build a boat should be followed or else you risk getting something wrong.*

process *noun* a series of actions or steps taken in order to achieve a particular end: *Military operations could jeopardise the peace **process**.*

pulley *noun* a piece of equipment consisting of a wheel over which a rope or chain is pulled to lift heavy things: ***Pulley** systems are used when there is a need to lift heavy objects.*

pump *noun* a machine used to raise or move fluids in a specified direction: *The water would not reach the third floor without a **pump** to lift it.*

Q

qualified *adjective* professional: *He is a **qualified** mechanic.*

R

radiation *noun* energy from heat or light that you cannot see: *Exposure to high **radiation** can lead to serious health problems.*

radio waves *noun* an electromagnetic wave used for long-distance communication: ***Radio waves** can travel for thousands of miles.*

radiograph *noun* an image produced on a sensitive plate or film by X-rays, gamma rays or similar radiation and typically used in medical examination: *A **tooth radiograph** can give a very detailed image of the tooth.*

radiographer *noun* a healthcare professional who specialises in the imaging of human anatomy to check for potential diseases: ***Radiographers** who develop X-ray films are exposed to many dangerous chemicals that put them at risk of developing asthma among other health issues.*

radiography *noun* the process or occupation of taking radiographs to assist in medical examinations: ***Radiography** is used in airport security.*

RAM *noun* Random-Access Memory; the main memory in a computer where the programs and data in current use are kept: *There are new types of **RAM** currently under development, which can preserve data while powered down.*

rapidly *adverb* happening in a short time or at a fast pace: *The country's economy is growing **rapidly**.*

Raytheon's XOS 2 Exoskeleton *noun* a robotic suit that allows the user to lift 200 pounds hundreds of times without exhaustion: ***Raytheon's XOS 2 Exoskeleton** was named one of the best inventions of 2010.*

reasonable *adjective* fair; sensible: *These items have a **reasonable** price.*

regulation *noun* the ability of a power supply to maintain an output voltage within a specified tolerance as referenced to changing conditions of input voltage: *Many factors affect voltage **regulation**, including the cable size and type.*

renewable *adjective* energy that replaces itself naturally, or is easily replaced because there is a large supply of it: *Water, wind and solar power are examples of **renewable** energy.*

repair *verb* to fix: *I tried to **repair** the broken chair myself.*

repetitive *adjective* characterised by repetition: *This song has a **repetitive** beat.*

reservoir *noun* a receptacle or chamber for storing a fluid: *The mechanic checked the **reservoir** to detect where the leak was coming from.*

response *noun* reaction: *The alarm goes off in **response** to a possible threat.*

reverse *verb* to move backwards: *He was having trouble **reversing** his car to park.*

robotic *adjective* relating to robots: *Some people who lost their arms are equipped with a **robotic** arm that functions the same way as a human arm.*

run *verb* **1** to operate: *The car **runs** on unleaded fuel.* **2** to flow in a particular direction or place: *Water was **running** off the roof.* **3** to organise or be in charge of an activity, business, organisation or country: *Courses are currently being **run** in London and Edinburgh.* **4** to move very quickly, by moving your legs more quickly than when you walk: *Several people **ran** to help her when she fell.*

S

scan *verb* to examine or look for viruses in a computer: Tarek **scanned** the computer for any possible viruses.

scoot *verb* to move quickly and suddenly: They **scooted** off on their bikes.

seminar *noun* a conference or other meeting for discussion or training: A **seminar** is relatively more informal than a lecture.

sensor *noun* a device that detects and measures a physical property and records, indicates and responds to it: Some places have **sensors** on their doors to detect the presence of people in order to open.

server *noun* a computer or computer program that manages access to a centralised resource or service on a network: A game **server** allows several computers or gaming devices to play multiplayer games.

sewer *noun* a pipe or passage under the ground that carries away waste material: The water company was called to fix the blocked **sewer**.

shaft *noun* a long column or vertical passage: The cable which holds up the cabin runs all the way through the **shaft**.

shower head *noun* a perforated pipe or tube for spraying water on the person taking a shower: My mum bought a new **shower head** to match the taps.

shut down *phrasal verb* to turn off: You should **shut down** the computer before you unplug the wire.

shutter *noun* a device that opens and closes to expose the film in a camera: **Shutter** speed is the amount of time a camera shutter is open exposing light onto the camera sensor.

significantly *adverb* in an important way or to an important degree: The temperature has increased **significantly** these past couple of days.

simultaneously *adverb* occurring, operating or done at the same time: Two letters from the library arrived **simultaneously**.

socket *noun* a point in the wall or a piece of equipment where you can connect to a power supply: I plugged the washing machine's wire into the **socket** to get electricity.

software *noun* the programs and other operating information used by a computer: He needs to update his **software**.

solar *adjective* related to sun: **Solar** power uses the sun as its source of energy.

solder *verb* to join or repair metal surfaces with solder: The new wires have been **soldered** on to the loom.

soldering iron *noun* a tool which is heated, usually by electricity, and used for melting solder and putting it onto surfaces: Before you start, make sure that you have a **soldering iron** with a suitable tip to hand.

sophisticated *adjective* (of a machine) cleverly designed and working in a complicated way: The new design of this machine has a very **sophisticated** system.

specialist *noun* someone who knows a lot about a particular subject and is very skilled at it: This man is a **specialist** in repairing computers.

speed gun *noun* a device used to measure the speed of moving objects: The police uses **speed guns** to find out if a driver has been driving above the speed limit.

speedometer *noun* an instrument on a vehicle's dashboard indicating its speed: The **speedometer** was indicating 60 km/hr.

spindle *noun* a part of a machine, such as a CD player, shaped like a stick, around which something turns: *He replaced the broken **spindle** on the record.*

SSD *noun* a storage device used in place of a hard disk because of its much greater speed: *The demand for higher input/output performance has led to the development of **SSD's**.*

stabilise *verb* to make stable and immovable: *The air conditioner helps **stabilise** the temperature in the room.*

stain *noun* **1** a special liquid that you use to change the colour of something, esp. wood: *He sells a wide variety of **stains** for wood.* **2** a mark that is difficult to remove, esp. one made by a liquid such as coffee or ink: *There was a dark **stain** on the sofa.*

stain-repellent *adjective* the ability of a fabric to resist wetting and staining by water: *I need to buy some **stain-repellent** fabric to make a new jacket.*

stationary *adjective* standing still instead of moving: *How did you manage to move into a **stationary** vehicle?*

steam engine *noun* an engine that uses the expansion or rapid condensation of steam to generate power: *The **steam engine** left many people unemployed when it was first invented.*

steer *verb* to control and direct: *The driver **steers** the bus into the correct direction.*

suit *noun* a set of clothes to be worn on a particular occasion or for a particular activity: *I bought my brother a new business **suit** for his birthday.*

switch on *phrasal verb* to turn on a machine, light, radio, etc. using a switch: *He **switched on** the torch.*

synchronous *adjective* existing or occurring at the same time: *Their arrival to the finish line was **synchronous**.*

T

technician *noun* a person who works with a certain technical skill: *The **technician** was the only one who could fix the broken refrigerator.*

telecom *noun* telecommunications; communication over a distance by cable, telegraph, telephone or broadcasting: ***Telecom** is the abbreviated term for telecommunication.*

telescopic sight *noun* an optical sighting device that is based on a telescope that changes direction when it comes to an angle: *A **telescopic sight** is one part of a speed gun.*

thermal *adjective* related to heat: ***Thermal** energy is produced from high temperatures.*

thermometer *noun* a tool used to measure temperature: *The **thermometer** read ten degrees Celsius yesterday.*

thump *noun* the dull sound that is made when something hits a surface: *The book she was holding fell to the ground with a **thump**.*

traction *adjective* the act of pulling: *A **traction** lift is based on a pulley system.*

transformer *noun* a device for transforming or increasing the voltage of an alternating current: *There are three types of potential **transformers**, including electromagnetic, capacitor and optical.*

tread-walk *noun* kinematic walking machine with active velocity control: *The walking patterns of a **tread-walk** are different from the treadmill's.*

trigger *noun* something that causes something else to happen: *The **trigger** has detected the location of the car.*

trigger *verb* to cause something to happen very quickly: *This strong earthquake **triggered** a tsunami.*

tumour *noun* a group of diseased cells in your body that have divided and increased too quickly: *The doctor discovered a **tumour** in her brain.*

turbine *noun* an electrical mill or fan-like machine: *The **turbine** turns water to produce power.*

U

ultrasonic *adjective* describing sound waves that are too high for humans to hear: ***Ultrasonic** devices can detect objects and measure distances.*

undetected *adjective* not discovered: *The intruder's presence in the house was **undetected**.*

unwell *adjective* ill: *She did not go to work today because she was feeling **unwell**.*

update *verb* to make something more modern or up to date: *Security measures are continually **updated** and improved.*

USB *noun* Universal Serial Bus; a connection technology that can be attached to a computer, providing fast data exchange: *He safely removed his **USB** from the computer's port.*

V

vaccine *noun* a substance used to stimulate the production of antibodies and provide immunity against one or several diseases: *In the past, there were no **vaccines** against flu.*

Visually Impaired Assistant (VIA) *noun* a device aiming to help people who can't see find a simpler way to get around: *Designer Noam Kloppe has introduced the **Visually Impaired Assistant (VIA)**.*

vital *adjective* important: *Refrigeration is **vital** in preserving food.*

voltage *noun* electrical force measured in an electrical current: *A voltmeter is an instrument that measures **voltage**.*

W

wavelength *noun* the distance between two waves of energy such as sound or light: *The **wavelengths** of sound frequencies audible to the human ear measure between 20Hz and 20Khz.*

X

X-ray *noun* a photographic or digital image of the internal composition of something, especially a part of the body: *The **X-ray** showed that his arm was broken.*

حقوق التوزيع في الجمهورية العربية السورية
محفوظة للمؤسسة العامة للطباعة

تدقيق ومواءمة كتب اللغة الإنكليزية

لينا الذياب	رشا بايزيد
نبال حنا	سرور شلش
باسل صادق	شهرزاد الفراتي
علا داوود آغا	سمر الشيشكلي



322 Old Brompton Road,
London SW5 9JH,
England

Maktabat El Nashr El Tarbawi El Souri
(Syrian Educational Publishers)

Omar El Mukhtar 2nd Str., Bldg. 6
El Mazraa, Damascus-Syria
Phone: (011) 44676789
Fax: (011) 44676788
e-mail: info@syrianep.com
www.syrianep.com

© York Press 2018

*All rights reserved; no part of this publication
may be reproduced, stored in a retrieval system,
or transmitted in any form or by any means, electronic,
mechanical, photocopying, recording, or otherwise,
without the prior written permission of the Publishers.*